

Fig. 7: PCB design generation

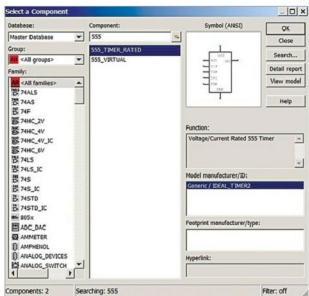


Fig. 8: Component window of Multisim

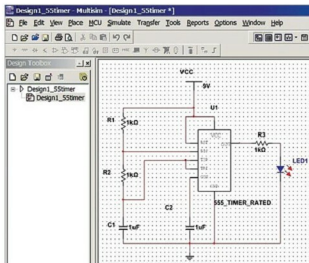


Fig. 9: Astable-multivibrator simulation in Multisim

artwork. For that, go to File menu, click on Mirror PCB artwork in Print option.

Multisim

Multisim is one of the most-commonly-used EDA tools in industries. It supports drawing and simulation of electronic circuits. It provides schematic capture and simulation application of National Instruments Circuit Design Suite, a suite of EDA. Multisim is designed for schematic entry, simulation and feeding to downstream steps such as PCB layout. It also includes mixed analogue/digital simulation capability and microcontroller (MCU) co-simulation.

Ultiboard is used to design PCBs, perform certain basic mechanical CAD operations and prepare these for manufacturing. It also provides automated parts placement and layout. The same circuit used for illustrating the previous section will be used here for familiarising with this software.

Follow the steps given below for drawing and simulating the circuit. Here we use Multisim 12.0 and Ultiboard 12.0 for simulation.

1. Double-click on Multisim icon on the desktop or in the program list. The icon can also be found in the installed folder (normally in C:/Program Files/National Instruments/Circuit Design Suite xx/).

2. As in other EDA tools, a schematic window will open along with a navigation window.

3. Next, place the components in the schematic window by clicking Place→Components. This will open a new component window (Fig. 8).

4. From the component window, select the required components. Multisim supports most of the components present in the

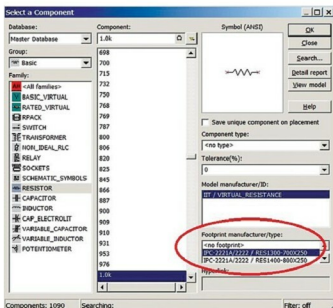


Fig. 10: Footprint recognition